Continuous Random Variables and Distributions

Xiaojing Ye

Department of Mathematics and Statistics

Georgia State University

Continuous Random Variables

A continuous RV X can take any value in an interval of $\mathbb{R}$.

Therefore, the set of possible values of X is uncountable, and it is not meaningful to consider $P(X = x)$ for any specific x.

Instead, we only consider $P(a < X < b)$ for some interval (a, b).

Distribution Functions

We call p the **probability density function** (pdf) of X if

$$P(a < X < b) = \int_a^b p(x)\, dx$$

for **any interval** (a, b) **in** $\mathbb{R}$. Note a, b can be $\pm\infty$.

Due to the property of integral, we see

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$$

We define F as the **cumulative distribution function** (cdf) of X if

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} p(z)\, dz$$

for any $x \in \mathbb{R}$.

Properties of pdf

- **Non-negative:** $p(x) \geq 0$.
- **Unity:** $\int_{-\infty}^{\infty} p(x)dx = 1$.

Properties of cdf

- F is non-decreasing.
- (Continuous RV only) F is differentiable and

$$p(x) = \frac{d}{dx}F(x)$$

- $0 \leq F(x) \leq 1$ for all $x \in \mathbb{R}$.
- The trend of F:

$$\lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to +\infty} F(x) = 1$$

Example. Suppose a continuous RV has pdf

$$p(x) = \begin{cases} \frac{k}{x^3}, & x \geq 1 \\ 0, & x < 1 \end{cases}$$

Find the value of k, the cdf F of X, and $P(X > 5)$.

Solution. For unity, we need

$$1 = \int_{-\infty}^{\infty} p(x)dx = \int_1^{\infty} \frac{k}{x^3} dx = \frac{k}{2}.$$

Hence $k = 2$. Clearly $F(x) = P(X \leq x) = 0$ if $x < 1$. For $x \geq 1$, there is

$$F(x) = \int_{-\infty}^{x} p(z)dz = \int_1^{x} \frac{2}{z^3} dz = 1 - \frac{1}{x^2}.$$

Moreover, $P(X > 5) = 1 - P(X \leq 5) = 1 - F(5) = 1 - (1 - \frac{1}{5^2}) = 0.04$.

Expectation of continuous RVs

The **expectation** of a continuous RV X with pdf p is defined by

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x p(x) \, dx$$

- Shorthand notations: μ_X or μ.
- Expectation is a **weighted integration** of all possible values.
- The weight of each point x is $p(x)$.

Expectation of a function

Let X be a RV and $Y = g(X)$ for some function g. Then

$$\mathbb{E}(Y) = \mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)p(x)\, dx$$

Variance and Standard Deviation

The **variance** of X is defined by

$$\mathrm{Var}(X) = \mathbb{E}\big((X-\mu)^2\big) = \int_{-\infty}^{\infty} (x-\mu)^2 p(x)\, dx$$

Shorthand notations of variance: σ_X^2 or σ^2. We still can show $\mathrm{Var}(X) = \mathbb{E}(X^2) - \mu^2$.

The **standard deviation** of X is

$$\sigma_X = \sqrt{\mathrm{Var}(X)}$$

Sometimes also written as $\mathrm{std}(X)$.

Remark. $\mathrm{Var}(X)$ is always non-negative.

Covariance and Correlation between two RVs

The **covariance** between X and Y is defined by

$$\mathrm{Cov}(X, Y) = \mathbb{E}\big((X - \mu_X)(Y - \mu_Y)\big) = \mathbb{E}(XY) - \mu_X \mu_Y$$

The **correlation** between X and Y is defined by

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- Shorthand notation: $\rho_{X,Y}$ or ρ.
- Correlation is a **normalization** of covariance.
- $-1 \leq \mathrm{Corr}(X, Y) \leq 1$

All properties about expectation, variance, covariance, and correlation still hold as for discrete RVs:

- $\mathbb{E}(aX + bY + c) = a\mathbb{E}(X) + b\mathbb{E}(Y) + c$
- $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$
- $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
- $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
- $\mathrm{Cov}(aX + b, Y) = a\mathrm{Cov}(X, Y)$
- $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + 2\mathrm{Cov}(X, Y) + \mathrm{Var}(Y)$

If X and Y are **independent**, then

- $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.
- $\mathrm{Cov}(X, Y) = 0$.
- $\mathrm{Corr}(X, Y) = 0$.

Warning: Neither of these implies X and Y are independent in general!

Uniform distribution

Let $a < b$ be real numbers. We say $X \sim \mathrm{Uniform}(a, b)$ if its pdf is

$$p(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a < x < b, \\ 0, & \text{elsewhere.} \end{cases}$$

We find

$$\mathbb{E}(X) = \int_a^b xp(x)dx = \int_a^b \frac{x}{b-a}dx = \frac{a+b}{2}$$

$$\mathbb{E}(X^2) = \int_a^b x^2p(x)dx = \int_a^b \frac{x^2}{b-a}dx = \frac{a^2+ab+b^2}{3}$$

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{(b-a)^2}{12}$$

Affine transform of uniform RV

If $X \sim \mathrm{Uniform}(a, b)$ and

$$U = \frac{X - a}{b - a} \quad \text{or equivalently} \quad X = a + (b - a)U,$$

then $U \sim \mathrm{Uniform}(0, 1)$.

We call U the standard uniform RV. Notice that

$$\mathbb{E}(U) = \frac{1}{2}, \qquad \mathrm{Var}(U) = \frac{1}{12}$$

Exponential distribution

We say $X \sim \mathrm{Exponential}(\lambda)$ if its pdf is

$$p(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

We can find its cdf is

$$F(x) = \begin{cases} 1 - e^{-\lambda x}, & \text{if } x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Remark. Exponential distribution is often used to model waiting time, failure time, etc.

Suppose $X \sim \mathrm{Exponential}(\lambda)$. Using integration by parts, we find

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} xp(x)dx = \int_{0}^{\infty} x\lambda e^{-\lambda x}dx = \frac{1}{\lambda}$$

$$\mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2p(x)dx = \int_{0}^{\infty} x^2\lambda e^{-\lambda x}dx = \frac{2}{\lambda^2}$$

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{1}{\lambda^2}$$

Connection between $\mathrm{Exponential}(\lambda)$ and $\mathrm{Poisson}(\lambda)$

Fix a unit time period, and suppose a certain event occurs independently for λ times on average during this period. Let Y be the number of events during this time period. We know $Y \sim \mathrm{Poisson}(\lambda)$.

Then the time between any two consecutive events follows $\mathrm{Exponential}(\lambda)$.

Proof sketch: Let $X_1, X_2, \cdots \overset{\mathrm{iid}}{\sim} \mathrm{Exponential}(\lambda)$ be the time gaps between two events, and $Y \sim \mathrm{Poisson}(\lambda)$ be the number of events. Then we just need to show for any $n = 0, 1, \ldots$ there is

$$P(Y = n) = P\left(\sum_{i=1}^{n+1} X_i > 1 \text{ and } \sum_{i=1}^{n} X_i \leq 1\right)$$

Note the joint pdf of $X_1, \ldots, X_n$ is $\lambda^n e^{-\lambda \sum_{i=1}^n x_i}$ for all $n \geq 1$ and $x_i > 0$.

To show this equation, we calculate the right-hand-side: We just need to integrate the joint pdf of $(X_1, \ldots, X_{n+1})$ over the domain

$$\Big(\sum_{i=1}^{n+1} x_i > 1\Big) \bigcap \Big(\sum_{i=1}^{n} x_i \leq 1\Big) \subset \mathbb{R}_+^{n+1},$$

which yields

$$\int_{0 \leq \sum_{i=1}^n x_i \leq 1} \lambda^n e^{-\lambda \sum_{i=1}^n x_i} \Big(\int_{1-\sum_{i=1}^n x_i}^{\infty} \lambda e^{-\lambda x_{n+1}} dx_{n+1}\Big)(dx_n \cdots dx_1)$$

After integration, we find this is $\frac{\lambda^n e^{-\lambda}}{n!}$, which is exactly the left-hand-side $P(Y = n)$.

Example. A printer receives on average 3 jobs per hour. Find the expected time between two consecutive jobs and the probability that a new job will be received in 5 minutes.

Solution. Let X be the number of jobs received in 1 hour. Then $X \sim \mathrm{Poisson}(3)$. Let Y be the time between two consecutive jobs, then $Y \sim \mathrm{Exponential}(3)$. Therefore

$$\mathbb{E}(Y) = \frac{1}{3}.$$

Since 5 minutes is $\frac{1}{12}$ hour, we get

$$P\left(Y \leq \frac{1}{12}\right) = F_Y\left(\frac{1}{12}\right) = 1 - e^{-3 \cdot \frac{1}{12}} = 1 - e^{-\frac{1}{4}}$$

where F_Y is the cdf of Y.

Gamma distribution

We say $X \sim \text{Gamma}(\alpha, \lambda)$ with shape $\alpha > 0$ and frequency $\lambda > 0$ if it has pdf

$$p(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \quad \text{for all } x > 0$$

where Γ is called the **Gamma function** and defined by

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$$

for any $\alpha \geq 1$.

Properties of Gamma function Γ

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$$

- Γ ensures the unity of p:

$$\int_0^\infty \lambda^\alpha x^{\alpha-1} e^{-\lambda x} dx = \int_0^\infty (\lambda x)^{\alpha-1} e^{-(\lambda x)} d(\lambda x) = \int_0^\infty t^{\alpha-1} e^{-t} dt = \Gamma(\alpha)$$

- For any $\alpha \geq 1$, there is

$$\Gamma(\alpha+1) = \int_0^\infty t^\alpha e^{-t} dt = -t^\alpha e^{-t}\Big|_0^\infty + \alpha \int_0^\infty t^{\alpha-1} e^{-t} dt = \alpha\Gamma(\alpha)$$

- Easy to see $\Gamma(1) = 1$. By the last property, there is $\Gamma(n) = (n-1)!$ for all $n > 1$.

Expectation and Variance of $\mathrm{Gamma}(\alpha, \lambda)$

Suppose $X \sim \mathrm{Gamma}(\alpha, \lambda)$. We find

$$\mathbb{E}(X) = \int_0^\infty x p(x) dx = \int_0^\infty x \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \frac{\Gamma(\alpha+1)}{\lambda\Gamma(\alpha)} \int_0^\infty \frac{\lambda^{\alpha+1}}{\Gamma(\alpha+1)} x^{(\alpha+1)-1} e^{-\lambda x} dx = \frac{\alpha}{\lambda}$$

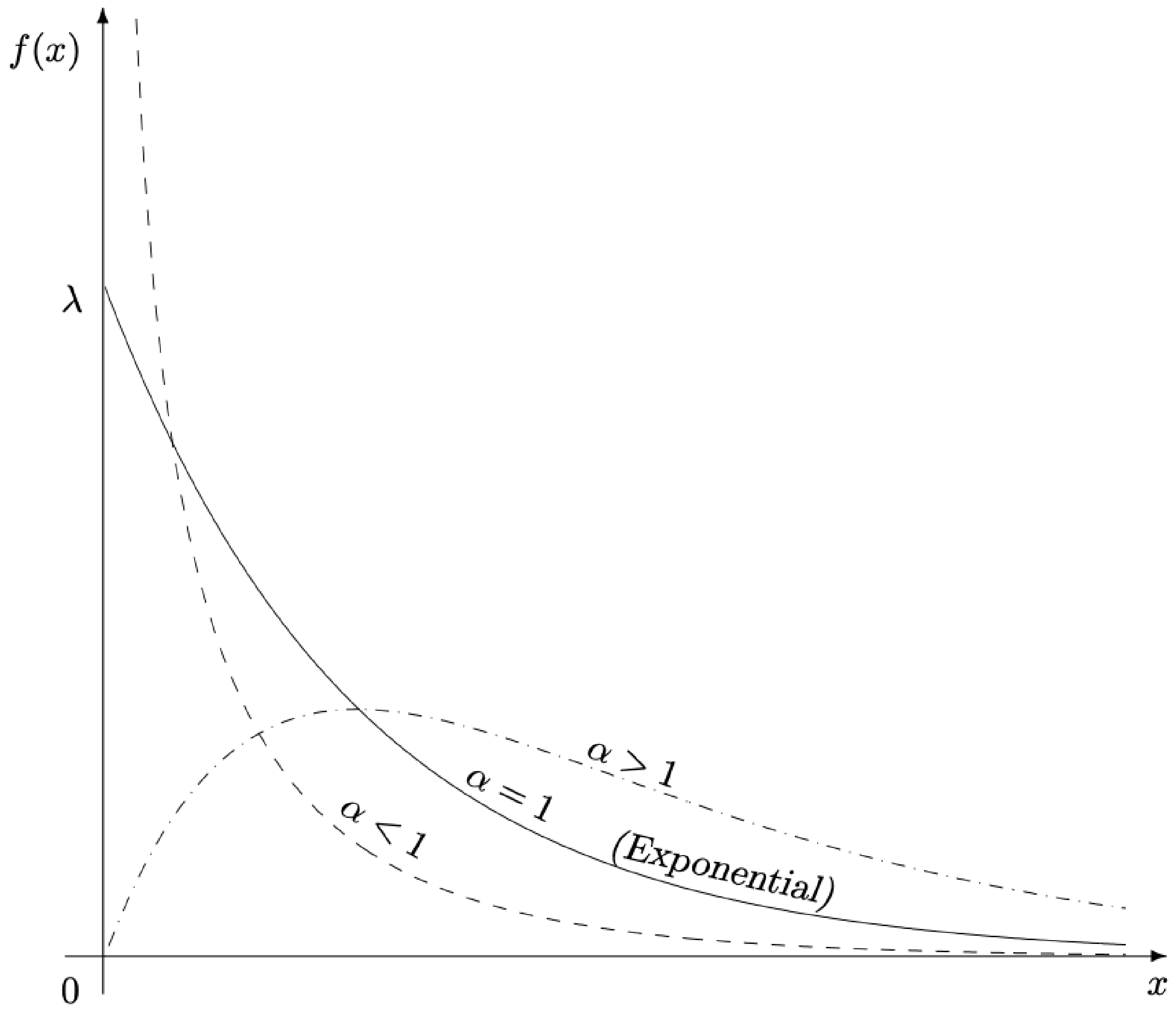
We also find

$$\mathbb{E}(X^2) = \int_0^\infty x^2 p(x) dx = \int_0^\infty x \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \frac{\Gamma(\alpha+2)}{\lambda^2\Gamma(\alpha)} \int_0^\infty \frac{\lambda^{\alpha+2}}{\Gamma(\alpha+2)} x^{(\alpha+2)-1} e^{-\lambda x} dx = \frac{\alpha(\alpha+1)}{\lambda^2}$$

Therefore

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{\alpha(\alpha+1)}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}$$

pdf of Gamma(α, λ) with a fixed frequency λ and varying shape α



Moment Generating Function

Let X be a RV. The **moment generating function** (MGF) of X is defined by

$$G_X(t) = \mathbb{E}(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} p(x) dx$$

for any $t > 0$. The name came from the fact that

$$\frac{d^k}{dt^k} G_X(t) \Big|_{t=0} = \mathbb{E}(X^k)$$

for all $k = 1, \ldots,$, where $\mathbb{E}(X^k)$ is called the k**-th moment of** X.

Remarks.

- Two RVs with the same MGF have identical p.
- If X and Y are independent, then $G_{X+Y}(t) = G_X(t) G_Y(t)$.

Claim. If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exponential}(\lambda)$, then

$$\sum_{i=1}^{n} X_i \sim \text{Gamma}(n, \lambda)$$

Proof. Notice the MGF of $X_i \sim \text{Exponential}(\lambda)$ is

$$G_{X_i}(t) = \mathbb{E}(e^{tX_i}) = \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - t}$$

Since $X_1, \ldots, X_n$ are independent, we get

$$G_{\sum_{i=1}^n X_i}(t) = \prod_{i=1}^{n} G_{X_i}(t) = \frac{\lambda^n}{(\lambda - t)^n}$$

which is exactly the MGF of $\text{Gamma}(n, \lambda)$.

Connection between Gamma and Poisson distributions

For any $n \in \mathbb{N}$, $\lambda > 0$, and $t > 0$, let

$$T \sim \text{Gamma}(n, \lambda), \qquad X \sim \text{Poisson}(\lambda t)$$

Suppose an event occurs at frequency λ and consider a time period of length λt. Then

$$\begin{aligned}(T > t) &= \text{It takes longer than } t \text{ to see } n \text{ events} \\ (X < n) &= \text{There are fewer than } n \text{ events occur during } \lambda t\end{aligned}$$

These two cases are identical! Hence $(T > t) = (X < n)$ and

$$P(T > t) = P(X < n) = F_X(n-1) \quad \text{and} \quad P(T \leq t) = P(X \geq n) = F_X(n)$$

where F_X is the cdf of $X \sim \text{Poisson}(\lambda t)$.

Remark. This shows the relation between the cdf of Poisson and Gamma distributions, but be careful about the choices of n, λ, t when using it.

Two special cases of Gamma distribution

- $\mathrm{Gamma}(1, \lambda) = \mathrm{Exponential}(\lambda)$
- $\mathrm{Gamma}(\frac{\nu}{2}, \frac{1}{2})$ is called the **chi-square distribution with degrees of freedom** ν, denoted by χ^2_ν.
  - If $Y = \sum_{i=1}^n Z_i^2$ where $Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, then $Y \sim \chi^2_n$.
  - We will introduce the Gaussian $\mathcal{N}$ next, and more use of χ^2_ν in Chapter 9.

Gaussian distribution (aka Normal distribution)

We say $X \sim \mathcal{N}(\mu, \sigma^2)$ if it has pdf

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

for all $x \in \mathbb{R}$.

Next we show

$$\mathbb{E}(X) = \mu, \qquad \mathrm{Var}(X) = \sigma^2$$

Expectation of $\mathcal{N}(\mu, \sigma^2)$

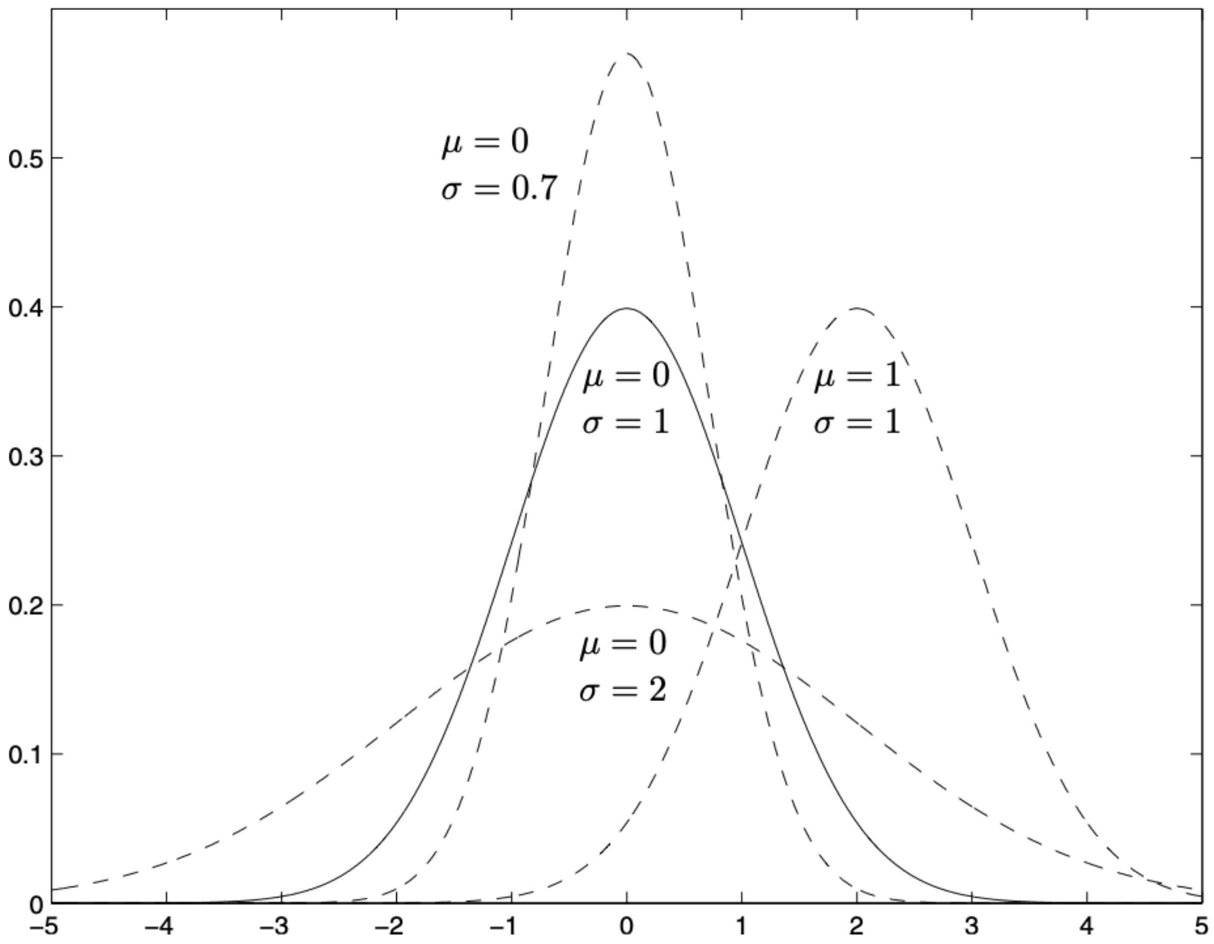
$$
\begin{aligned}
\mathbb{E}(X) &= \int_{-\infty}^{\infty} xp(x)dx = \int_{-\infty}^{\infty} (x-\mu+\mu)\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}dx \\
&= \int_{-\infty}^{\infty} (x-\mu)\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}dx + \mu\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}dx \\
&= \sigma\int_{-\infty}^{\infty} \underbrace{z\frac{1}{\sqrt{2\pi}}e^{-\frac{z^2}{2}}}_{\text{odd function}}dz + \mu\int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}}_{\text{pdf of }\mathcal{N}(\mu,\sigma^2)}dx \\
&= 0 + \mu \cdot 1 \\
&= \mu
\end{aligned}
$$

where the change of variable $z = \frac{x-\mu}{\sigma}$ is applied in the 3rd line.

Variance of $\mathcal{N}(\mu, \sigma^2)$

$$
\begin{aligned}
\mathrm{Var}(X) &= \int_{-\infty}^{\infty} (x-\mu)^2 p(x) dx = \int_{-\infty}^{\infty} (x-\mu)^2 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\
&= \int_{-\infty}^{\infty} \sigma^2 z^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = -\frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z \, de^{-\frac{z^2}{2}} \\
&= -\frac{\sigma^2}{\sqrt{2\pi}} \Bigg(\underbrace{ze^{-\frac{z^2}{2}} \Big|_{-\infty}^{\infty}}_{0} - \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz \Bigg) \\
&= \sigma^2 \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{\text{pdf of } \mathcal{N}(0,1)} \\
&= \sigma^2
\end{aligned}
$$

pdf of $\mathcal{N}(\mu, \sigma^2)$



Normalization of a Gaussian RV

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Z = \frac{X-\mu}{\sigma}$, then $Z \sim \mathcal{N}(0,1)$.

Z is called the standard Gaussian RV. Its pdf is

$$p_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

(We can show the unity of p_Z and hence of any $\mathcal{N}(\mu, \sigma^2)$ pdf.)

The cdf of Z is by convention denoted as

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

The value table of Φ is easy to find.

Example. Suppose $X \sim \mathcal{N}(900, 200^2)$. Find $P(600 \leq X \leq 1200)$.

Solution. We apply a normalization $Z = \frac{X-900}{200}$ to get

$$\begin{aligned}
P(600 \leq X \leq 1200) &= P\left(\frac{600-900}{200} \leq \frac{X-900}{200} \leq \frac{1200-900}{200}\right) \\
&= P(-1.5 \leq Z \leq 1.5) \\
&= \Phi(1.5) - \Phi(-1.5) \\
&= \Phi(1.5) - (1 - \Phi(1.5)) \\
&= 2\Phi(1.5) - 1 \\
&\approx 0.8664
\end{aligned}$$

Example. Suppose $X \sim \mathcal{N}(900, 200^2)$. Find x such that $P(X \leq x) = 0.03$.

Solution. We apply a normalization $Z = \frac{X-900}{200}$ to get

$$
\begin{aligned}
0.03 &= P(X \leq x) \\
&= P\Big(\frac{X - 900}{200} \leq \frac{x - 900}{200}\Big) \\
&= P\Big(Z \leq \frac{x - 900}{200}\Big) \\
&= \Phi\Big(\frac{x - 900}{200}\Big)
\end{aligned}
$$

We find that $\frac{x-900}{200} = -1.88$ and hence $x = 524$.

Central Limit Theorem (CLT)

Let $X_1, \ldots, X_n$ be independent RVs (not necessarily the same distribution), all of which have expectation μ and variance σ^2. Let $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then

$$\mathbb{E}(\overline{X}) = \mu, \qquad \mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}.$$

Moreover, for any $z \in \mathbb{R}$,

$$P\Big(\frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \le z\Big) \to \Phi(z) \quad \text{as} \quad n \to \infty$$

This effectively implies that $\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$ approximately follows $\mathcal{N}(0,1)$ when n is large (typically $n \geq 30$).

Example. Suppose $X_1, \ldots, X_{300}$ are independent RVs, while they all have expectation 1 and standard deviation 0.5. Find the probability their sum is smaller than 330.

Solution. Let $\overline{X}$ be their average. Then $300\overline{X}$ is their sum. Then

$$\begin{aligned} P(300\overline{X} \leq 330) &= P\Big(\overline{X} \leq \frac{330}{300} = 1.1\Big) \\ &= P\Big(\frac{\overline{X} - 1}{0.5/\sqrt{300}} \leq \frac{1.1 - 1}{0.5/\sqrt{300}} = 2\sqrt{3}\Big) \\ &= P(Z \leq 2\sqrt{3}) \\ &= \Phi(2\sqrt{3}) \end{aligned}$$

where $Z = \frac{\overline{X}-1}{0.5/\sqrt{300}} \sim \mathcal{N}(0, 1)$ by CLT.

Gaussian approximation to Binomial distribution

Let $S \sim \text{Binomial}(n, \theta)$ with large n. We can regard

$$S = X_1 + \cdots + X_n, \quad \text{where} \quad X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$$

Then the sample mean $\overline{X} = \frac{S}{n}$ with $\mathbb{E}(\overline{X}) = \theta$ and $\text{Var}(\overline{X}) = \frac{\theta(1-\theta)}{n}$. As n is large, we have by CLT that

$$\frac{S - n\theta}{\sqrt{n\theta(1-\theta)}} = \frac{\overline{X} - \theta}{\sqrt{\theta(1-\theta)/n}} \sim \mathcal{N}(0, 1)$$

or equivalently $S \sim \mathcal{N}(n\theta, n\theta(1-\theta))$.

This implies $\text{Binomial}(n, \theta) \approx \mathcal{N}(n\theta, n\theta(1-\theta))$ when n is large.

Summary of continuous distributions:

Distribution	x values	pdf $p(x)$	$\mathbb{E}(X)$	$\mathrm{Var}(X)$
$\mathrm{Uniform}(a, b)$	(a, b)	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$\mathrm{Exponential}(\lambda)$	$(0, \infty)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$\mathrm{Gamma}(\alpha, \lambda)$	$(0, \infty)$	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
$\mathcal{N}(\mu, \sigma^2)$	$(-\infty, \infty)$	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2